Promo Finder: solution

1. What it does

Reads timestamped screenplays and returns ranked 20–60 s promos (segments, hook, cliffhanger, rationale, score, publishing plan) plus every rejected candidate with the reason. It only selects moments. See `README.md` and `CHANGELOG.md`.

2. How it works (five stages)

1. **Parse**: scripts become numbered blocks (dialogue, ACTION, SFX, CAMERA) with source times (3,727 blocks).
2. **Scout (LLM)**: proposes clips per episode, plus cross-episode ideas planned from the loglines. **It picks block numbers, never timestamps**; the tool copies times from the script, so timestamp hallucination is impossible by construction. It can still misdescribe what it picked, hence stages 3–5.
3. **Validate (code)**: 20–60 s; hook starts in the first 3 s; clip ends ≤ 4 s after the cliffhanger; hook and cliffhanger actually on screen; no dialogue cut mid-line; segments in story order. A failing clip is first **repaired by trimming its edges in code** (it keeps the model's hook and cliffhanger), and only then gets one LLM repair attempt. Every change is logged.
4. **Blind critic (LLM)**: a second prompt sees only the clip, as a first-time viewer would, and scores the 100-point rubric. It flags weak hook, missing context, predictable ending, low curiosity and misleading edit. Its scores replace the scout's own. Clear failures are rejected. **One weak subjective score on a clip totalling 75 or more goes to human review instead of rejection.** Reviews of identical clips are reused from the cache.
5. **Rank and review**: dedups, ranks, writes a publishing plan and runs local **human-review checks**. An **independent verifier** (own parser) then re-checks every timestamp, duration and score against the raw scripts.

3. Validation and the human in the loop

Free LLMs are imperfect, so the tool states its uncertainty. Each review flag names its evidence and suggests **accept / inspect / repair / reject**; subjective concerns never auto-reject, and an editor decides.

Kind	Checks	Basis
Objective	hook late/off screen; cliffhanger off screen; camera-only hook; split sentence; story order; displayed vs source times; missing or zero critic score	exact, from blocks
Approximate	hook/cliffhanger description not supported by the clip	word overlap; can only ask for a look
Subjective	critic flags; low coherence; scout vs critic disagree by ≥ 15 points	LLM judgement

`audit` re-checks a finished run and `compare` scores any runs with one method; both make **no model calls**. Six known good/bad clips check the critic (v3 reuses earlier scores for them). 60 tests cover broken clips (weak hooks, misleading endings, missing context, mid-line cuts), tampered outputs, quota errors and cache accounting.

4. Results and the recall study

The raw output holds v3, the final run. v2 (first version) and v4 (an experiment) were development runs, shown only for comparison: same models (free Gemini, batch mode) and match rule (a promo covers ≥ 50 % of a reference).

	v2 (first version)	v3 (final, in raw output)	v4 (experiment)
Promos: recommended / needs review	31: 14 / 17	48: **17 / 31**	54: 18 / 36
Reference recall, all promos	8 / 17 (47 %)	**13 / 17 (76 %)**	12 / 17 (71 %)
Reference recall, recommended only	5 / 17	5 / 17	4 / 17
Critic probes correct	6 / 6*	6 / 6	6 / 6

Why v2 missed 9 references: all 9 *had been proposed*, then lost: the LLM repair returned the same broken range (R02, R13), one low subjective score rejected an 81–82/100 clip (R07, R05), or a cross-episode setup was never paired (R08). **v3** fixes 27 of 30 invalid clips by trimming in code and sends one-weakness clips to review instead of rejecting them (4 live requests). Recall over all promos rose 47 % → 76 %, but every new match arrives as "needs review". Of the 19 new promos, 9 are good, 5 borderline and 5 poor, so the quality-adjusted gain is +3, not +5. **v3 is the final result:** the clean list grew (14 → 17) and weaker additions are labelled (`RECALL_STUDY.md` in the raw output).

v4 (9 live requests) re-scouts the 15 episodes without a clip: 7 new promos (3 good, 3 borderline, 1 poor marked recommended), 40 episodes. It lost R03 to a dedup rule since fixed; replayed with the fix it equals v3's recall with 19 recommended. Arc planner v2 still missed R08, so v4 was not adopted. *Probes and the 75-point rule were set after seeing data, and the references were picked by an AI, so recall is a consistency check, not ground truth.

5. Assumptions

Script times match the video; cuts fall on block boundaries (shots may be trimmed, lines never); the hook beat starts within 3 s; a new viewer knows only the clip; spoilers affect order, not score.

6. Models, free tier and scale

One `complete(system, prompt)` interface: Claude (default), Gemini, Groq, OpenRouter, Ollama, any OpenAI-compatible API, cache-only replay, or an offline heuristic; switching is one flag. For free tiers it paces requests, retries on HTTP 429, stops on the daily quota and resumes from its cache. `--batch` puts 10 episodes or 15 clips in a request (about 16 requests instead of 250).

7. Why results are limited, and how to improve them

All-promo recall is 76 % (47 % was v2), but **only 5 / 17 references are among the recommended clips**: most good finds wait in the review queue. v3's own output shows why:

- **The free models are imprecise.** 30 of 82 proposed clips broke a timing rule (code fixed 27), and 7 promos carry a hook or cliffhanger description their clip does not support.
- **Their judgement is noisy.** Of 31 promos sent to review, 15 have scout and critic scores ≥ 15 points apart and 11 one weak subjective score. Batching 10 episodes or 15 clips per request spreads attention and makes reviews less independent.
- **Arcs are planned from loglines**, so long-range pay-offs (R08) are missed; the 17 AI-picked references are a small, subjective yardstick; and text cannot show acting or pace.

A stronger model (e.g. Claude Opus 5.5: `--provider anthropic`, no code change) targets the model errors: more exact block choices, descriptions that match the clip, steadier scores (fewer disagreement and borderline flags), and, with a paid quota, one request per episode and per clip instead of batches. It cannot fix the text-only limit or the subjective references. **Not yet measured:** run the same command with `--provider anthropic`, then `compare` it with v3. **Other levers:** references and probes labelled by editors, to set thresholds on real judgement; `--per-episode 5` for more candidates; two critics that must agree; arc planning from full scripts; a check of the rendered video before publishing.

8. AI tools used, and where AI got it wrong

Claude Code (Claude Opus 5.5) read the scripts and wrote the code, tests, reference picks and docs; Gemini (free tier) runs the pipeline. Mistakes the system caught, all fixed and tested:

- **The independent verifier caught an AI-written validator** letting a promo end 0.2 s into the line *"Harrington! Gerald Harrington paid us!"*.
- **The audit found five promos whose cliffhanger shot is never on screen** (the validator counted trimmed shots); P010's description invented *"as beams collapse"*.
- **Gemini wrote `"narrative_ coherence"`**, read as 0/10 for P029; keys are now normalised. **Repairs often returned the same broken range**; code trimming now runs first.